

TUNKU ABDUL RAHMAN UNIVERSITY OF MANAGEMENT AND TECHNOLOGY

FACULTY OF ACCOUNTANCY, FINANCE AND BUSINESS

ACADEMIC YEAR 2025/2026

OCTOBER EXAMINATION

BAIT3003 DATA WAREHOUSE TECHNOLOGY

SATURDAY, 4 OCTOBER 2025

TIME: 2.00 PM – 4.00 PM (2 HOURS)

BACHELOR IN BUSINESS ANALYTICS (HONOURS)

Instructions to Candidates:

Answer **ALL** questions. All questions carry equal marks.

BAIT3003 DATA WAREHOUSE TECHNOLOGY

CUCKOO Malaysia: Wellness for All

Picture an ad of a slick machine that produces filtered water, hot or cold, by just clicking a button elegantly demonstrated by a Korean Oppa.

That is CUCKOO.

Today, other than water filters, CUCKOO Holdings Co., Ltd manufactures and exports a range of home appliances such as air purifiers, multi-cookers, and kitchen appliances to overseas markets worldwide. In Malaysia, Singapore, & Brunei, CUCKOO International (MAL) Bhd (CIM) is currently appointed as the sole importer and distributor of these products.

1. Products

CIM generates 99% of its total revenue from retailing CUCKOO products. The balance of 1% in total revenues was attributable to sales of beauty care, nutrition, and indoor disinfection products under brands such as WonderLab, WonderDewi, and WonderKlean and also provision of IT support and maintenance services. (The 1% in total revenue is classified under “Others”). There are four segments of CUCKOO products.

CUCKOO Branded – innovative home appliances and household goods:

There are 54 SKUs under its product portfolio, which comprises water filters, home appliances, kitchen appliances and air purifiers. Since 2021, its revenue contribution had dipped from 98.5% to 79.0% on 30 September 2024.

CUCKOO Co-Created – collaborations with home appliances and household goods industry leaders:

There are 10 SKUs in its product portfolio. These products are air-conditioners, massage chairs, and mattresses produced in partnership with Fujiaire, Ogawa and LSK. For this segment, revenue contribution increased from 2.9% in 2021 to 20.6% on 30 September 2024.

In this instance, CUCKOO-branded remains as its key revenue contributor in the present while it generates growing sales from its CUCKOO Co-Created products.

WonderLab & WonderDewi – beauty and wellness products

WonderKlean – professional home care and cleaning services

Year	CUCKOO-branded		CUCKOO Co-Created		Others	
	Revenue (RM '000)	Revenue (%)	Revenue (RM '000)	Revenue (%)	Revenue (RM '000)	Revenue (%)
2021	1,091,190	95.8%	32,928	2.9%	14,958	1.3%
2022	938,980	89.7%	95,353	9.1%	12,113	1.2%
2023	954,068	85.4%	153,859	13.8%	8,722	0.8%
9 Months Up to Sep 2024	734,700	79.0%	191,721	20.6%	3,944	0.4%

Table 1 Main revenue stream

BAIT3003 DATA WAREHOUSE TECHNOLOGY**2. Types of Revenue**

For both CUCKOO-branded and CUCKOO Co-Created products, CIM would generate five different types of revenue as follows:

- Outright Sales:** It refers to customers who pay full price for CUCKOO's products upfront.
- Finance Lease Sales:** It refers to customers who rent CUCKOO's products for a fixed monthly instalment for a period of 3-7 years depending on its type of products and sales packages. After the completion of their instalments, the ownership of products will be transferred to customers.
- Finance Lease Interest Income:** It refers to the return made by CIM for providing the finance lease facility to customers renting CUCKOO's products.
- Rental Income:** It refers to customers, who rent CUCKOO's products for monthly instalments, but had opted not to continue owning these products after completion of these instalments. This source of income had ceased beginning in October 2020. Hence, CUCKOO's rental income had dwindled down since 2021, reflecting its decision to cease offering such contracts to customers.
- Revenues from maintenance services** and sales of **spare parts** to its customers.
- Among them, CIM's revenues were generated substantially from finance lease sales & maintenance and spare part sales. It added margins from finance lease interest income.

Year	Outright Sales		Finance Lease Sales		Financial Lease Interest Income		Rental Income		Maintenance and Spare Parts	
	Sales in RM '000	Sales (%)	Sales in RM '000	Sales (%)	Sales in RM '000	Sales (%)	Sales in RM '000	Sales (%)	Sales in RM '000	Sales (%)
2021	60,749	5.3%	342,858	30.1%	72,106	6.3%	260,949	22.9%	387,456	34.0%
2022	73,741	7.1%	292,386	27.9%	41,751	3.9%	211,732	20.2%	414,723	40.0%
2023	55,577	4.9%	413,561	37.0%	49,127	4.4%	167,438	15.0%	422,224	37.0%
9M till Sep 24	35,631	3.8%	455,943	49.0%	42,380	4.6%	79,302	8.5%	313,165	33.7%

Table 2 Types of Revenue

3. Profit Margin

In 2021-2023, CIM generated RM 1.0-1.2 billion in revenues per annum. In this period, its adjusted net profits have declined from RM 186.4 million in 2021 to RM 100.6 million in 2023. This decline was attributed to a rise in labour costs, marketing costs, and net losses arising from impairment of finance lease and trade receivables during the period.

Year	2021	2022	2023	9M till Sep 2024
Revenue (RM '000)	1,139,076	1,046,446	1,116,649	930,365
Gross Profits (RM '000)	400,340	385,007	370,115	289,937
Gross Profit Margins (%)	35.1%	36.8%	33.1%	31.2%
Adjusted Net Profits (RM '000)	186,436	151,081	100,585	103,689
Adjusted Margins (%)	16.4%	14.4%	9.0%	11.1%
Adjusted Net Profits a Share (RM)	0.130	0.105	0.070	0.072

Table 3 Profit Margin

BAIT3003 DATA WAREHOUSE TECHNOLOGY**4. Cash Conversion Cycle**

CIM has a cash conversion cycle of 312 days. It is attributable to its high level of trade receivables. For the period ended 30 September 2024, its trade receivable turnover is 337 days, which indicates that CIM takes around 337 days to collect cash from customers. As for its inventory turnover, it is at 57 days, indicating that CIM takes around two months to sell off its inventories. Meanwhile, its trade payable turnover stands at 82 days, indicating that it took as long as 2-3 months to pay off its suppliers (mainly CUCKOO Holdings Co., Ltd).

	<u>FYE2021</u>	<u>FYE2022</u>	<u>FYE2023</u>	<u>FPE2024</u>
Trade receivables turnover period (days) ⁽¹⁾	217	283	307	337
Trade payables turnover period (days) ⁽²⁾	150	76	88	82
Inventory turnover (days) ⁽³⁾	49	67	62	57

Table 4 Cash conversion cycle

Expansion Plans

CIM plans to allocate RM5.0 million to open **10 CUCKOO “cash and carry” brandshops** in the next two years which consist of 1,000 sq. ft. in size per store and would feature CUCKOO’s products. In 2025, it plans to open three brandshops in Kuala Lumpur, Penang and Johor. In 2026, it wishes to open seven brandshops in Selangor, Perak, Kedah, Kelantan, Terengganu, and Sabah.

Furthermore, it will spend RM10.0 million in **expanding** its operations in **Singapore**, which mostly consists of advertising and promotion expenses.

CIM is also allocating RM5.6 million in **upgrading** its **IT system**, particularly its warehouse management system.

Latest Partnership

On June 23rd 2025, Cuckoo International (MAL) Bhd (CIM) and Samsung Malaysia Electronics have entered a strategic partnership to roll out a new suite of smart healthy home solutions in Malaysia.

The move is set to expand Cuckoo's product lineup to include Samsung's consumer electronics.

The collaboration, named the Cuckoo Smart Healthy Home Co-Created Programme with Samsung, marks Cuckoo's latest push under its "co-created" business segment, which has become a key revenue driver for the company. Under the partnership, Cuckoo will distribute Samsung products including smartphones, smart TVs, AirDressers, refrigerators, and washing machines, integrating them into its wellness-focused product ecosystem. The new range of products will be available for purchase beginning July 1.

Business risks moving forward:

CIM’s business model would need a substantial amount of cash upfront to finance the purchase of products from its suppliers (mainly, CUCKOO Holdings Co., Ltd) and pay sales commissions. Subsequently, CIM is dependent on its customers for prompt payment of their monthly instalments for a longer period of time.

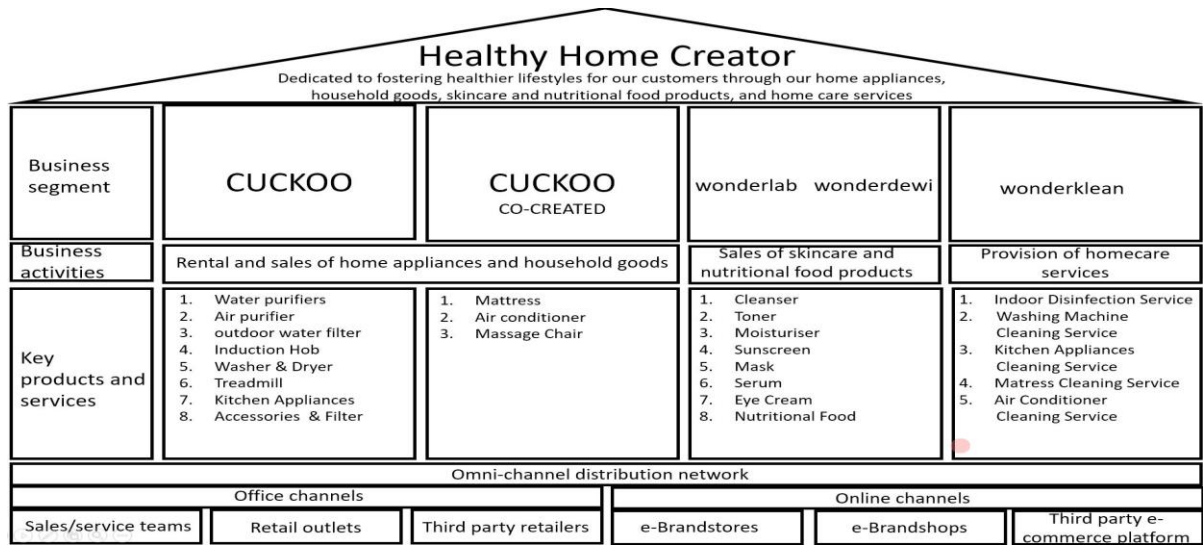
Adapted from:

1. Bursa: 8 Things to Know About CUCKOO MY Before Subscribing to its IPO Shares – KCLau.com. (n.d.). Retrieved June 25, 2025, from <https://kclau.com/investment/8-things-to-know-about-cuckoo-my-before-subscribing-to-its-ipo-shares/>

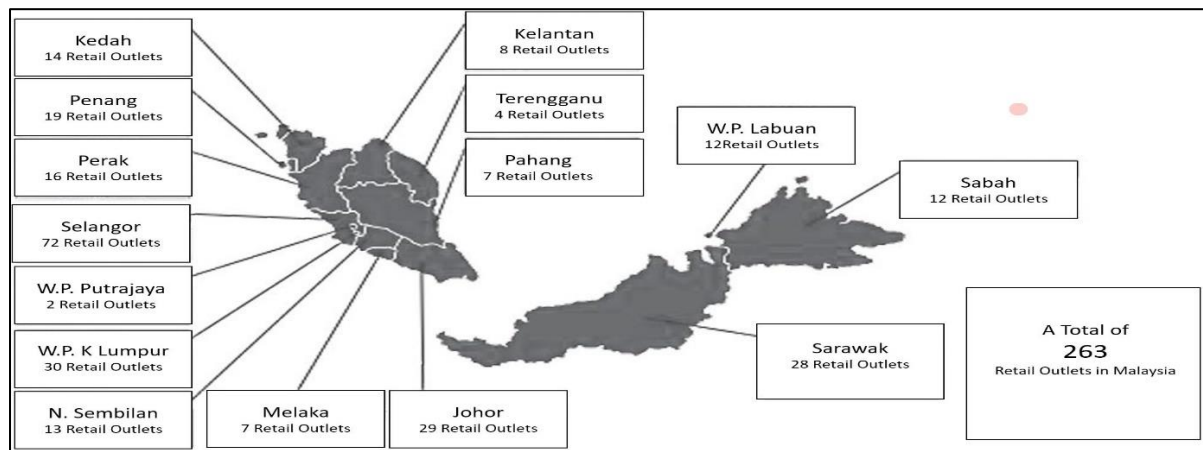
2. Cuckoo Malaysia’s Bursa Listing: A Defining Moment For Consumer Wellness. (n.d.). <https://www.Businesstoday.Com.My/>. Retrieved June 25, 2025, from <https://www.businesstoday.com.my/2025/06/22/cuckoo-malaysias-bursa-listing-a-defining-moment-for-consumer-wellness/>

3. NOTICE ACCOMPANYING THE ELECTRONIC PROSPECTUS OF CUCKOO INTERNATIONAL (MAL). (n.d.). Retrieved June 25, 2025, from www.cuckoo.com.my

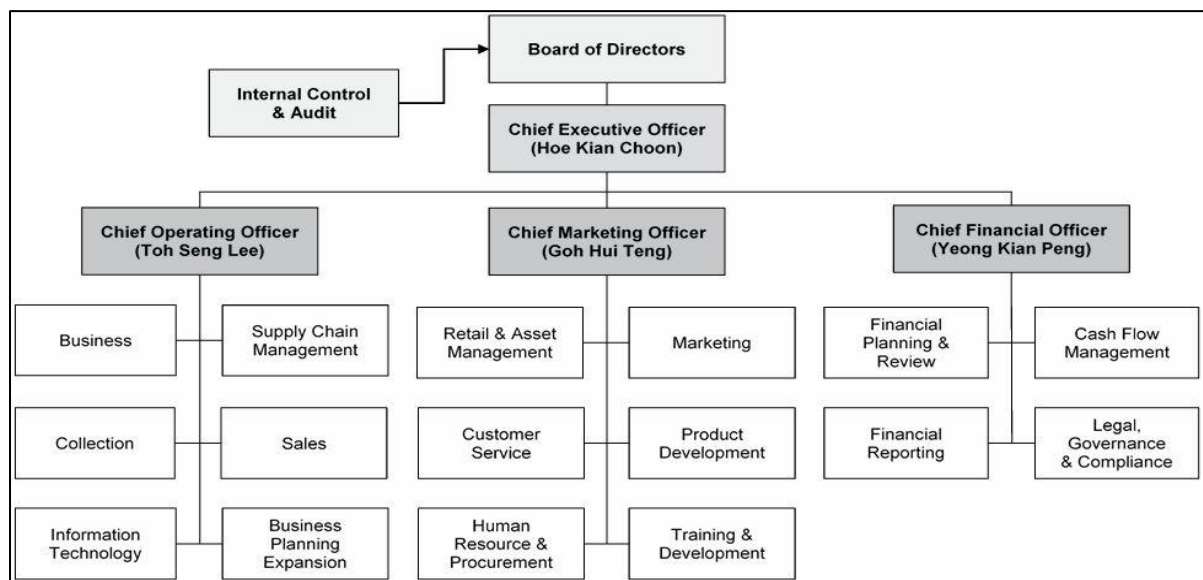
BAIT3003 DATA WAREHOUSE TECHNOLOGY



Summary of CUCKOO business



Locations in Malaysia



CUCKOO Malaysia Management (Business Function)

BAIT3003 DATA WAREHOUSE TECHNOLOGY**Question 1**

- a) CIM currently has an IT system to manage its daily transactions. CIM is considering to create a data warehouse (DWH) system to assist in its strategic planning and operations.

Explain why the DWH should ideally at least be a 3-tier architecture. (10 marks)

- b) CIM is considering to implement a data warehouse based on a centralised architecture.

(i) Do you think this is a good choice? Provide relevant justification. (5 marks)

(ii) Briefly describe the architecture of a centralised data warehouse and explain the major components. (10 marks)

[Total: 25 marks]

Question 2

- a) Building a DWH can be costly and time consuming. Some members of the senior management is asking why not just use the existing OLTP system for business intelligence reporting.

Explain why having a data warehouse is more suitable for business analytics than just relying on the current system. (10 marks)

- b) In a Top-down approach, we would build a DWH first and then perform ETL to create data marts. In a Bottom-up approach, we first create data marts and then consolidate them to create a DWH.

Discuss with relevant examples/justifications, which approach is more advantages to CIM for its next 3 years of operation. Illustrate with a diagram and discuss in more detail how CIM would build the data warehouse/data mart using the selected approach. (15 marks)

[Total: 25 marks]

Question 3

- a) Suggest a Star-schema design for CIM's data mart/warehouse. Your design should have 4 dimension tables and 1 fact table. Populate your tables with relevant attributes according to the business case. (15 marks)

- b) To minimise start-up cost, CIM rents retail space for its outlets. CIM would like to see the trend in rental expenditure for each of its outlets. This information will help in budgeting for operational expenditure. Illustrate how this can be implemented with a 'Type 2 Slowly Changing Dimension (SCD)'. Provide relevant example(s). (10 marks)

[Total: 25 marks]

BAIT3003 DATA WAREHOUSE TECHNOLOGY**Question 4**

- a) CIM is now considering to set up an e-commerce platform to increase the sales of its packaged products. Assume that CIM would like to embark on a marketing and promotion campaign, with the main objective of increasing customer enrolment to its e-commerce platform. Explain how a data warehouse can maximise the information potential to assist in achieving CIM's objectives. (15 marks)
- b) Explain how a **dashboard** uses multidimensional data model to support data analysis and present the findings in various information formats. (10 marks)

[Total: 25 marks]